

Market Design for Generative AI: Beyond the Copyright Binary

<https://arxiv.org/abs/2606.12260>

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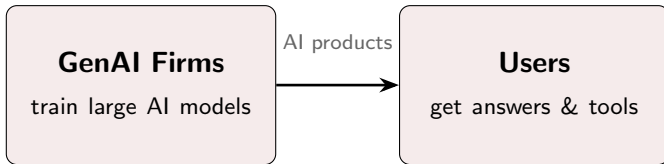
GenAI delivers fast and cheap content, but is laden with both promise and peril

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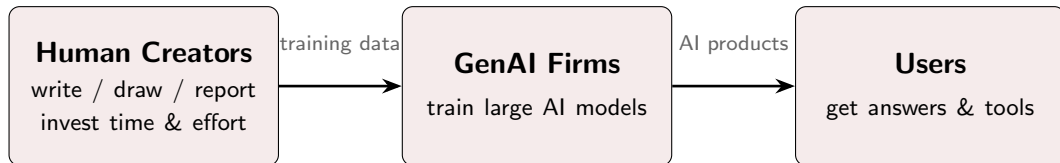
GenAI Firms

train large AI models

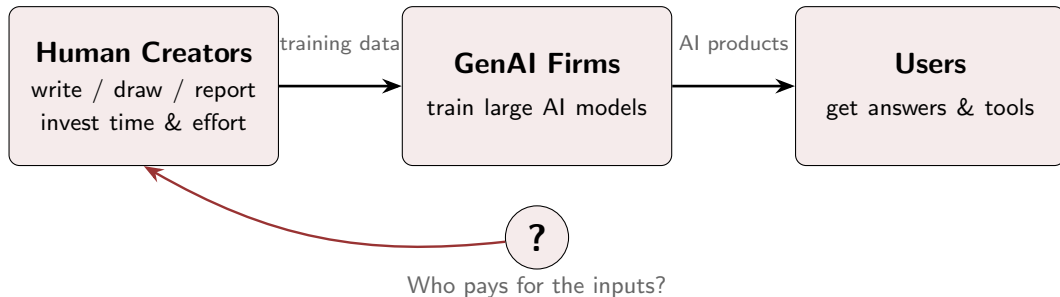
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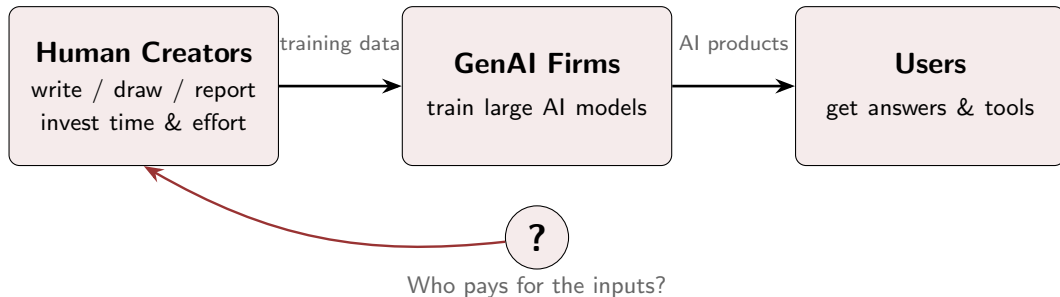
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A data market for human content that:

(i) enables technological progress and (ii) preserves incentives to create

Why this matters to MSOM?

Both creators and AI firms need a sustainable data market

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For AI product builders

- Data access drives model quality
- Legal risk becomes product risk
- Licensing costs & delays matter

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For creators and publishers

- AI (already!) diverts traffic & revenue
- Incentives shape what gets created
- Less willing to be “original” (soon...)

The copyright binary (and why it fails)

Free-for-all

“fair use”

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“fair use”

price to creators = 0

⇒ “free rider” problem

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Full IP rights

“infringement”

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Both extremes can under-deliver on long-run quality

Need “something in between”

Our theoretical model

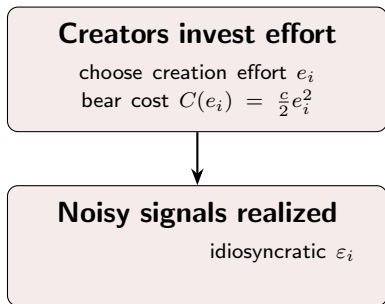
Creators invest effort

choose creation effort e_i

bear cost $C(e_i) = \frac{c}{2}e_i^2$

content is a noisy signal of “ground-truth” X

Our theoretical model



content is a noisy signal of “ground-truth” X

$$\varepsilon_1 \sim \mathcal{N}(0, e_1^{-1})$$

$$\varepsilon_2 \sim \mathcal{N}(0, e_2^{-1})$$

$$\varepsilon_N \sim \mathcal{N}(0, e_N^{-1})$$

Our theoretical model

Creators invest effort

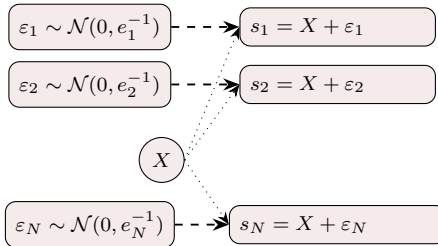
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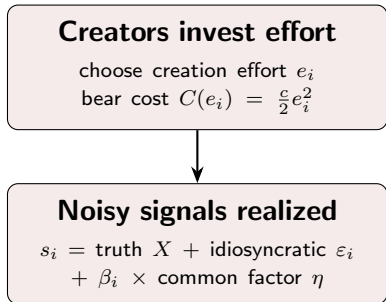
Noisy signals realized

$s_i = \text{truth } X + \text{idiosyncratic } \varepsilon_i$

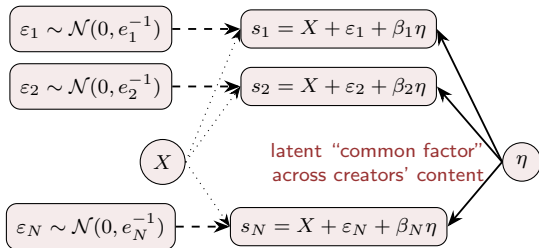
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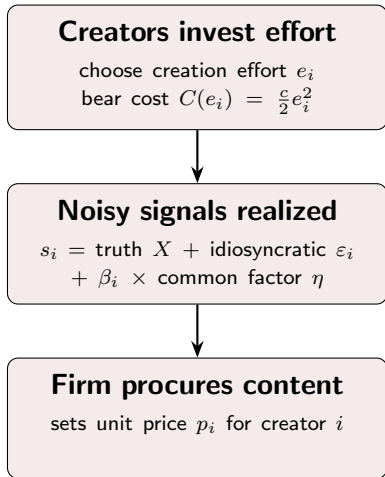
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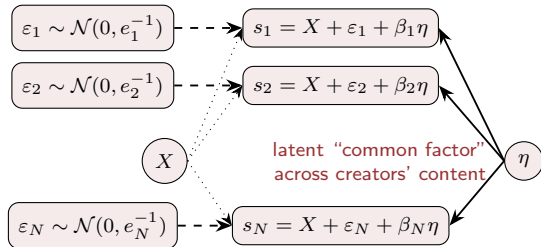
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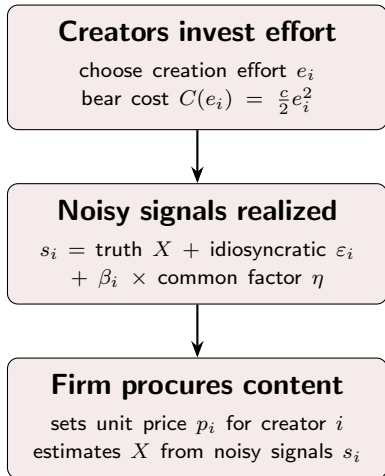


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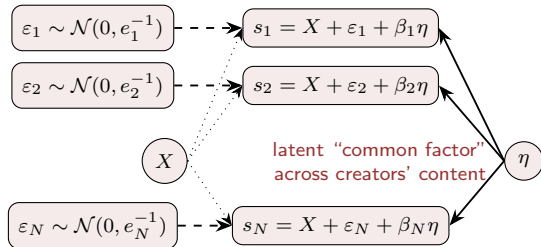


price $p_i \implies$ best response $e_i = p_i/c$

Our theoretical model



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price $p_i \implies$ best response $e_i = p_i/c$

service quality $Q := \text{Var}(\hat{X})^{-1}$ of BLUE $\hat{X}$

profit $\Pi := Q - \sum_i p_i e_i = Q - \sum_i c e_i^2$

Benchmark: First-best (social optimum) effort

First-best solution by a social planner

e^{FB} maximizes social welfare $W := Q - \sum_i \frac{c}{2} e_i^2$ (cf. firm's profit $\Pi = Q - \sum_i c e_i^2$)

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Regime 1: Free-for-all (“fair use”)

No payment

$$p_i = 0$$



No effort

$$e_i = 0$$



Bad AI model
precision collapse

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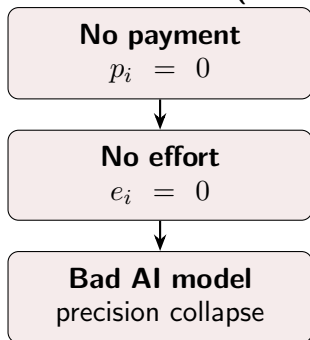
Regime 2: Full IP rights

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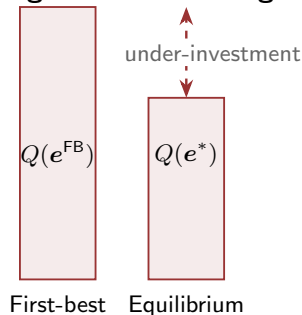
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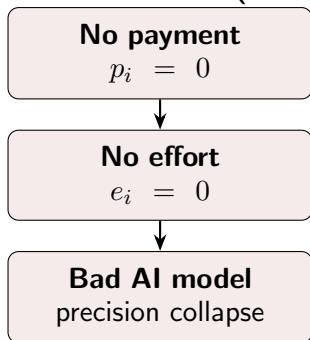


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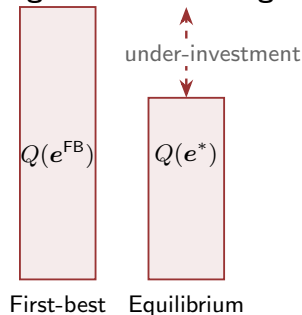
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But who suffers the most?

Static market failure: an “originality penalty”

Innovators vs. Imitators

- **Innovators:** unaffected by trends ($\beta_i \approx 0$).
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----- first-best

Does scarcity command a premium?

____ 0%
Innovators Imitators

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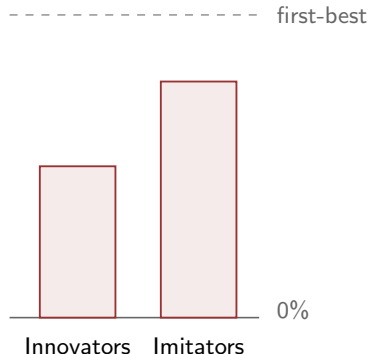
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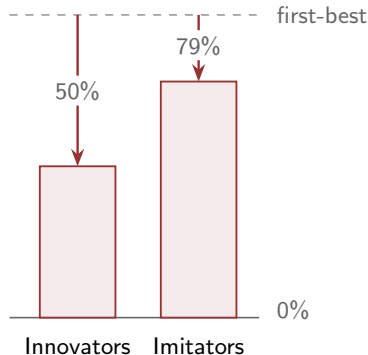
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$$R_i := \frac{e_i^*}{e_i^{\text{FB}}} \approx \begin{cases} 50\%, & \text{for innovator} \\ 79\%, & \text{for imitator} \end{cases}$$

and R_i is strictly increasing in β_i



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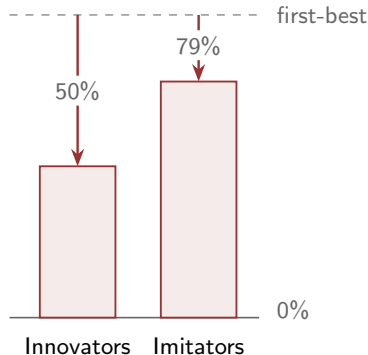
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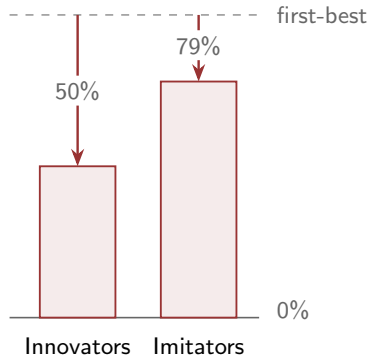
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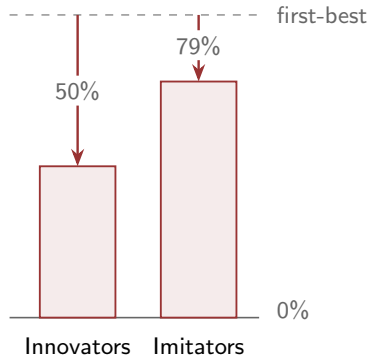
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- **Caveat.** 79% appears in a homogeneous market full of imitators. In some cases $R_i > 1$, i.e., $e_i^* > e_i^{\text{FB}}$!

Dynamic market failure: a “curse of precision”

A continuous-time fluid model with time $t \in [0, \infty)$

- Market price $p(t)$ draws in a mass $N_O(t)$ of innovators & $N_C(t)$ of imitators

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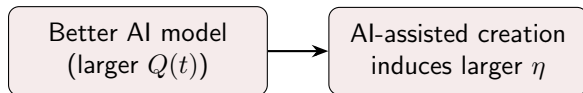
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Better AI model
(larger $Q(t)$)

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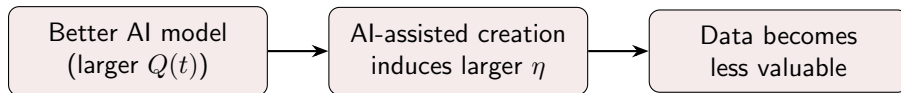
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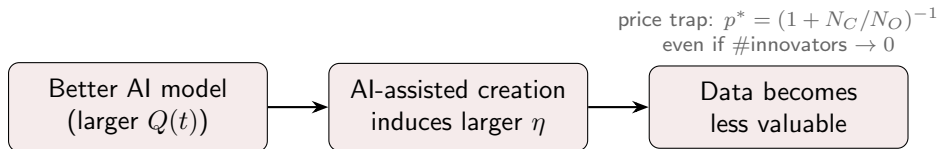
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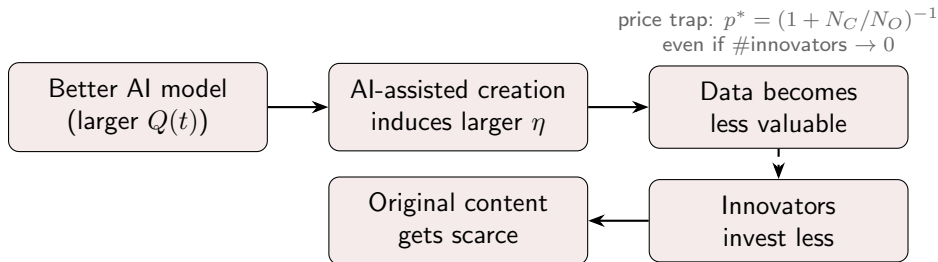
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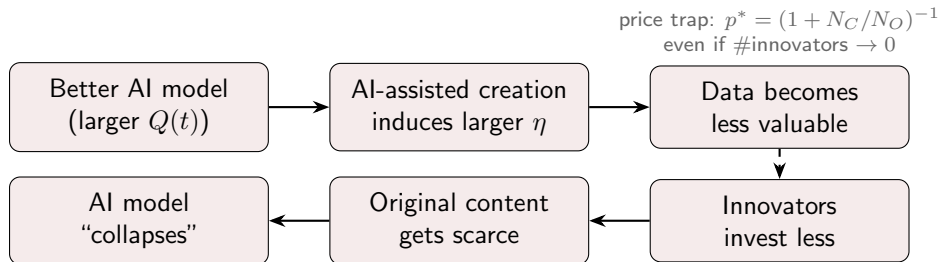
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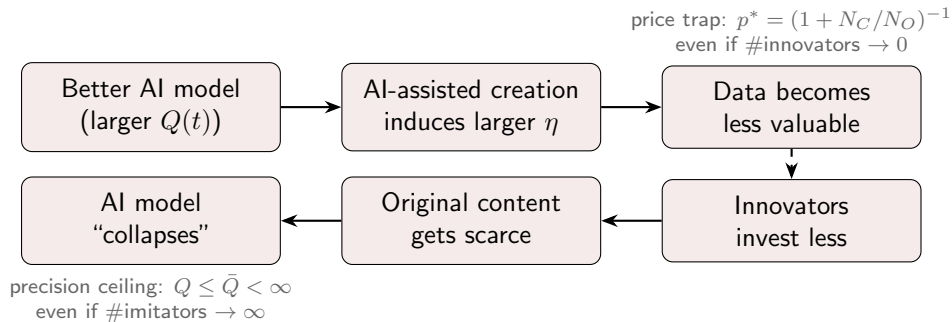
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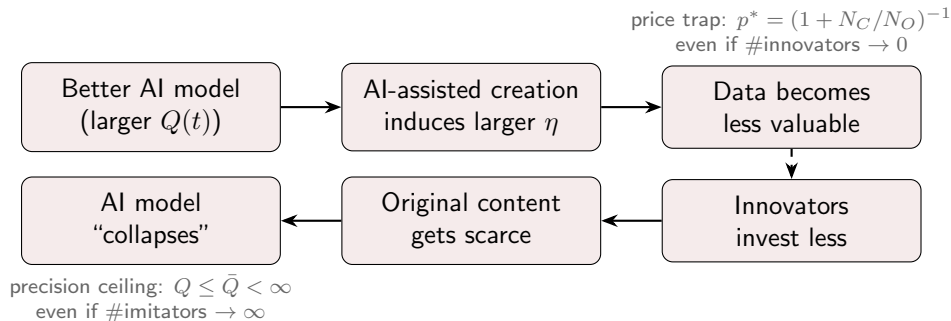
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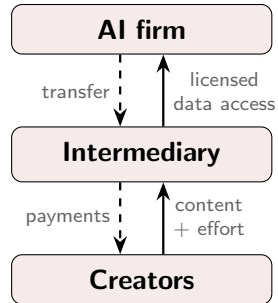
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Takeaway. Even with constant input of human data, “model collapse” still happens!

Toward a solution: a data intermediary

- Negotiates **collectively** to counter monopsony power
- Rewards **originality** across human creators

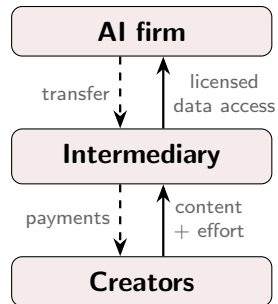


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What the intermediary offers

- To AI firms: licensed data access
- To creators: a fair share of surplus
- To society: implements the first best

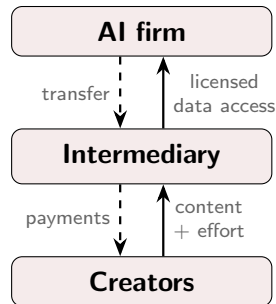


Toward a solution: a data intermediary with a two-part tariff mechanism

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$$\text{Payment } P_i(e) = \underbrace{c e_i^{\text{FB}}}_{\text{unit price}} \times e_i + \underbrace{B_i}_{\text{subsidy}}; \text{ Transfer } T(e) = \sum_{i=1}^N c e_i^{\text{FB}} \times e_i + \underbrace{L}_{\text{lump-sum}}$$

- Unit price $c e_i^{\text{FB}}$ induces best response $e_i^{\text{FB}} \implies$ Incentive Compatibility (IC)
- L via Nash bargaining, B_i via Aumann–Shapley $\implies$ Individual Rationality (IR)
- Intermediary bears neither deficit nor surplus $\implies$ Budget Balance (BB)

Main Takeaways

- In Generative AI data market, creators' content is **inherently correlated**

Thank you!

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Main Takeaways

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- AI-assisted content creation amplifies the failure to a dynamic **curse of precision**
- We propose a **data intermediary** utilizing two-part tariff mechanisms as a solution

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